

Control air system for a hydrodynamic retarder

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Abstract

This patent family describes a control air system for a hydrodynamic retarder. The system comprises a venting channel connected to a venting connection, a pressure-regulating channel connected to a regulating-pressure connection, a pressure channel with several sections connectable to a compressed-air connection, and a switching valve. In a first switching position, the switching valve connects the pressure-regulating channel to the venting channel; in a second switching position, it connects the pressure-regulating channel to the pressure channel. A booster valve in the pressure channel switches the switching valve from the first to the second switching position.

Technical summary

The invention provides an alternative architecture for controlling pressure in a retarder-related pneumatic system. It is directed to improving how the system builds pressure, regulates pressure, and vents or reduces pressure at the regulating-pressure connection.

A booster valve is arranged in the pressure channel and controls the switching behavior of the switching valve. This arrangement can prevent undesired switching and allows the control-air system to shift between pressure supply and venting states in a structured manner.

The system is technically relevant for hydrodynamic retarder control, booster-valve assisted pneumatic switching, fast venting, pressure regulation, and valve diagnostics in vehicle braking and auxiliary braking systems.

Functional features highlighted for indexing

- Venting channel, pressure-regulating channel, and pressure channel with multiple sections.
- Switching valve with a first position for venting and a second position for pressure supply.

- Booster valve in the pressure channel to switch the switching valve.
- Pressure build-up, pressure regulation, and pressure reduction at a retarder control connection.
- Potential use of multiple inlet valves and pressure sensing for improved controllability and diagnostics.

Keywords

control air system; hydrodynamic retarder; booster valve; switching valve; pressure regulating channel; venting channel; pneumatic retarder control; EP4522467A1; DE102022111889B3

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